Name: _____

WORKSHEET

Straight-line motion 1

1 Find the velocity and acceleration (ignore units) of a particle if its displacement is given by

a $s = 10t$

b $s = t^2$

c $s = 7t^2$

d $s = 3t^2 - 4t + 1$

e $s = t^3$

f $s = -2t^3$

g $s = -5t^2 + 4t^2 + t - 2$

h $s = t^4 - 10t^2$

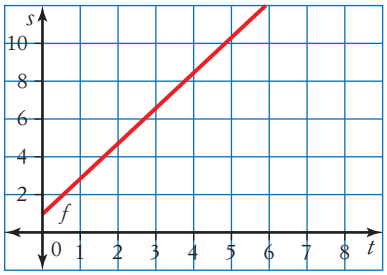
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i $s = -t^5 - 3t^4 + 20t^3 + 12t$

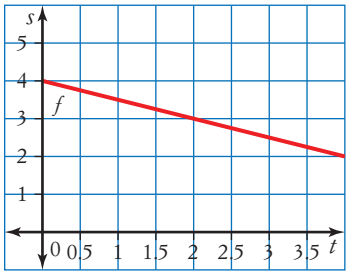
j $s(t) = ut + \frac{1}{2}at^2$

2 Given the following displacement time graphs, describe the nature of the velocity.

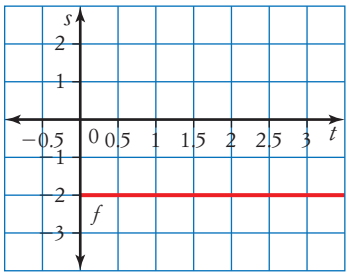
a



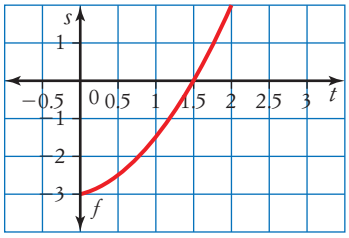
b



c

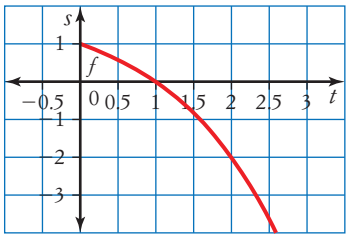


d

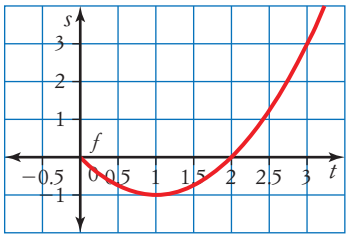


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e

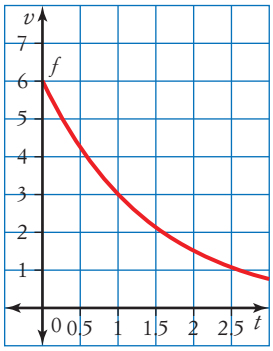


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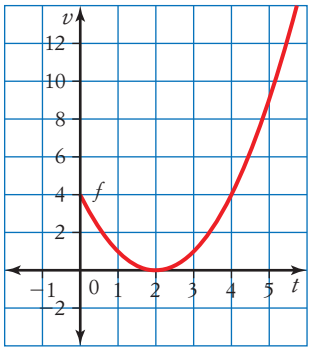


3 Given the following velocity time graphs, describe the nature of the acceleration.

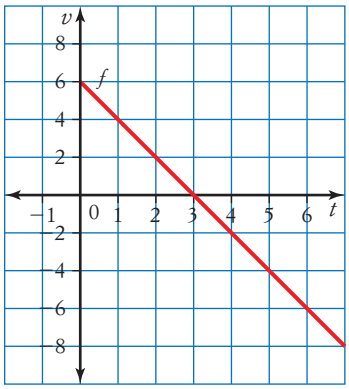
a



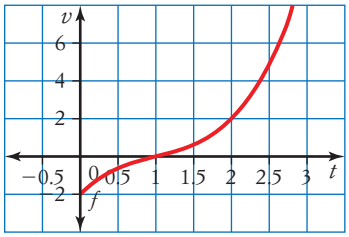
b



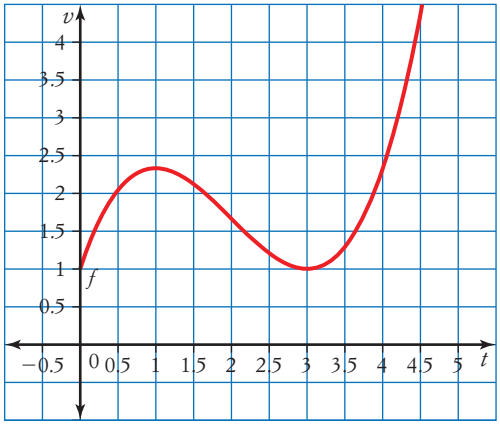
c



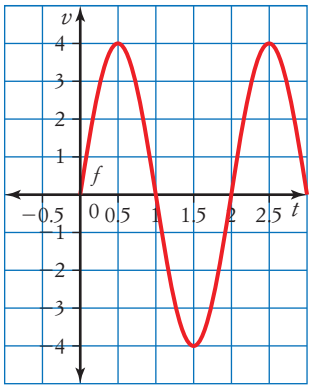
d



e



f



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Answers

1 a $v = 10, a = 0$

b $v = 2t, a = 2$

c $v = 14t, a = 14$

d $v = 6t - 4, a = 6$

e $v = 3t^2, a = 6t$

f $v = -6t^2, a = -12t$

g $v = -15t^2 + 8t + 1, a = -30t + 8$

h $v = -4t^3 - 20t, a = \text{c} -20$

i $v = -5t^4 - 12t^3 + 60t^2 + 12,\ a = -20t^3 - 36t^2 + 120t$

j $v(t) = u + at, a(t) = a$

2 a Velocity is constant and positive for all $t \geq 0$

b velocity is constant and negative for all $t \geq 0$

c $v = 0\ \forall\ t \geq 0$

d velocity is positive and increasing for all $t \geq 0$

e velocity is negative and increasing for all $t \geq 0$

f velocity is negative and decreasing for $0 \leq t \leq 1$

velocity $= 0$ for $t = 1$

velocity is positive and increasing for $t > 1$

3 a acceleration is negative and decreasing for all $t \geq 0$

b acceleration is negative and decreasing for $0 \leq t \leq 2$

acceleration $= 0$ for $t = 2$

acceleration is positive and increasing for $t > 2$

c acceleration is negative and constant for $t \geq 0$

d acceleration is positive and decreasing for $0 \leq t \leq 1$

acceleration is positive and increasing for $t > 1$

acceleration is positive at $t = 1$

e acceleration is positive and decreasing for $0 \leq t \leq 1$

acceleration $= 0$ for $t = 1, 3$

acceleration is negative and increasing for $1 < t < 2$

acceleration is negative and decreasing for $2 < t < 3$

acceleration is negative for $t = 2$

acceleration is positive and increasing for $t > 3$

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f acceleration is positive and decreasing for $0 < t < 0.5$, $2 < t < 2.5$, etc.

acceleration is positive and increasing for $1.5 < t < 2$, $3.5 < t < 4$, etc.

acceleration $= 0$ for $t = 0.5, 1.5, 2.5 \ldots$ etc.

acceleration is positive for $t = 0, 2$ etc

acceleration is negative for $t = 1, 3$ etc

acceleration is negative and increasing for $0.5 < t < 1$, $2.5 < t < 3$, etc.

acceleration is negative and decreasing for $1 < t < 1.5$ etc

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